AMSAVALLI R

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Email — LinkedIn — GitHub — Portfolio

PROFILE SUMMARY

Data Science and AI/ML Engineer with 3+ years of software engineering experience in developing Python applications, scalable REST APIs, data-driven systems, and AI/ML solutions. Proficient in Python, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, SQL, and MongoDB. Experienced in exploratory data analysis, data preprocessing, machine learning, deep learning, computer vision, model training, and evaluation. Hands-on experience building production-ready applications using FastAPI, Django, Docker, and cloud technologies.

TECHNICAL SKILLS

Programming Languages	Python, SQL, JavaScript, TypeScript
Data Science	Pandas, NumPy, Matplotlib, Exploratory Data Analysis, Data Cleaning, Data Preprocessing
Machine Learning	Scikit-learn, Regression, Classification, Random Forest, Model Training, Model Evaluation
Deep Learning	PyTorch, TensorFlow, CNNs, Transfer Learning, U-Net, Image Segmentation
Computer Vision	Image Analysis, Image Recognition, OCR, Image Preprocessing
Data Engineering	Data Pipelines, Data Processing, Hadoop, Spark, ETL
Databases	MySQL, PostgreSQL, MongoDB, Redis
AI / NLP	OpenAI, Natural Language Processing, Conversational AI, AI Agents
Backend Development	FastAPI, Django, Flask, REST APIs, Async Programming, WebSockets
Cloud / DevOps	AWS, Azure, Docker, Git, GitHub, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Python Developer, KiteCareer DEC 2023 – PRESENT

- Developed scalable Python applications and REST APIs using FastAPI and Django to support business workflows and data-driven applications.
- Designed and managed database operations using MySQL and MongoDB, including schema design, query optimization, and data processing.
- Improved database query performance by approximately 30% through query optimization and database schema improvements.
- Reduced API response times through backend optimization, efficient database operations, and improved application workflows.
- Developed data processing and transformation workflows using Python, handling structured data formats including JSON, CSV, and Excel.

DATA SCIENCE & MACHINE LEARNING PROJECT

CraveCall.ai– AI-Powered Voice Agent Live Website

Python, FastAPI, React.js, MongoDB, WebSocket, Twilio

Contributed to an AI-powered voice agent platform for automated customer calls, order placement, bookings, and appointment scheduling.

Integrated AI-driven workflows with backend APIs and real-time communication services to support automated customer interactions.

AI-Powered Matrimony Platform Live Website

Python, FastAPI, React.js, MongoDB, Stabilityai

Developed an AI-driven matrimony platform using FastAPI and MongoDB, planned as a SaaS-based backend system for intelligent matchmaking. Implemented machine learning models to analyze user profiles, preferences, and behavioral data to generate optimized match recommendations. Built secure REST APIs with RBAC-based authentication and authorization, enabling scalable and privacy-focused user management. Integrated analytics to track user engagement and improve recommendation accuracy, with a cloud-ready architecture aligned for AWS deployment and future containerization.

Social Media Automation Tool Live Website

Python, FastAPI, React.js, MongoDB,meta developer account

Developed a Social Media Automation Tool for Instagram and Facebook using FastAPI and MongoDB Atlas, enabling image uploads, scheduling, and direct publishing via the Meta Graph API. Integrated Cloudinary for optimized media storage and automated workflows with Make.com, delivering a scalable, API-driven architecture with multi-platform integration potential.

Profitkart Live Website

Python, FastAPI, MongoDB, REST APIs, JWT Authentication, Git, Linux, VPS Deployment

Developed a multi-vendor e-commerce platform connecting vendors and customers through a unified marketplace.

Implemented vendor management features including product management, order processing, employee permissions, delivery management, and subscription management.

Developed subscription-based product and order limits, add-on purchases, notifications, and reporting and analytics APIs.

Deployed backend services on a Linux VPS and managed application updates using Git.

Inclusive Innovative Street

Built an AI-powered urban analytics platform using machine learning, geospatial analytics, and optimization techniques to analyze street infrastructure. Applied data governance principles, analytics pipelines, and AI-driven optimization to identify safe routes and accessibility features. Leveraged OpenStreetMap data and geospatial registries and taxonomies for structured urban feature classification.

PYTHON DEPLOYMENT & DEVOPS EXPERIENCE

- Deployed Python applications using FastAPI and Django on Linux-based VPS environments.
- Managed Linux servers using SSH, command-line tools, and systemd services.
- Configured Nginx as a reverse proxy and managed Uvicorn application servers.
- Configured Python virtual environments, dependencies, environment variables, and production settings.
- Managed MongoDB and MySQL databases, including production database operations.
- Handled Git-based deployments, code updates, and production troubleshooting.
- Configured domains, SSL certificates, backend routing, and CORS settings.
- Monitored application logs and resolved server-side and deployment-related issues.

EDUCATION

MASTER OF COMPUTER APPLICATION 2020
Indira Gandhi National Open University - Tiruvandrum CGPA: 7.5

BACHELOR OF COMPUTER APPLICATION 2014
J.P College or Arts Science - Ayikudy CGPA: 7.9